

Normalization Type Predicts Regularization Response: An Optimizer-Conditional Three-Way Interaction in Vision Architectures

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Abstract

The optimal training recipe for a vision model depends critically on its architecture, yet practitioners routinely transfer recipes across architectures with unpredictable outcomes. We discover that this transfer instability is governed by a three-way interaction among normalization type, optimizer, and regularization strength. Through a systematic study of 10 architectures $\times$ 8 recipes $\times$ 3 seeds (up to 240 configurations) plus 3 causal norm-swap experiment pairs (72 additional runs), we show that under Stochastic Gradient Descent (SGD), Batch Normalization (BN) and Layer Normalization (LN) architectures are perfectly separated in their regularization response (Cliff’s $\delta=1.0$, $p=0.016$): every BN architecture benefits from regularization while every LN architecture is harmed. AdamW largely neutralizes this divide. The architecture-regularization interaction produces a large effect ($\eta_p^2=0.858$, $p<10^{-50}$), while regularization alone shows no main effect ($p=0.282$); the benefit depends entirely on the architecture. Norm-swap experiments causally confirm that changing only the normalization layer (e.g., replacing LN with BN in ConvNeXt) triggers catastrophic SGD collapse. These findings yield a zero-cost heuristic: inspecting a model’s normalization layer is sufficient to eliminate half the recipe search space before any training run.

1 Introduction

Apply the same regularization recipe to ResNet-18 and ConvNeXt-Tiny under SGD, and the outcomes diverge dramatically: ResNet-18 gains 7.63 percentage points in test accuracy while ConvNeXt-Tiny loses 11.75. The recipe is identical; only the architecture differs. This divergence is not a hyperparameter tuning failure. It is a fundamental interaction between architecture design and training recipe that current practice largely ignores.

Modern vision architectures ship with hand-tuned default recipes that differ substantially across families. ResNet variants are typically trained with SGD, moderate weight decay, and minimal data augmentation [25], whereas ConvNeXt and vision transformers adopt AdamW with aggressive regularization [12, 23]. When practitioners transfer a recipe from one architecture to another (as commonly happens when adapting published baselines to new settings), the results are unreliable. Yet no systematic framework explains *why* certain architectures require fundamentally different recipes or *which* architectural property drives the divergence.

We address this gap through a controlled factorial study: 10 diverse vision architectures trained under all $2 \times 2 \times 2 = 8$ combinations of optimizer (SGD, AdamW), data augmentation strength (light, strong), and regularization (none, standard), each repeated across 3 seeds. Complementing this correlational evidence, we conduct 3 targeted norm-swap experiments that surgically replace the normalization layer of an architecture while holding everything else fixed. The results converge on a single finding: **normalization type is the architectural property that determines regularization response, and this effect is conditioned on the optimizer**. Under SGD, BatchNorm architectures uniformly benefit from regularization while LayerNorm architectures are uniformly harmed; AdamW largely neutralizes this divide.

Three lines of evidence support this claim. First, causal norm-swap experiments confirm that changing only the normalization layer reverses or amplifies the regularization response; LN-to-BN swaps trigger catastrophic SGD collapse. Second, under SGD the BN/LN separation in regularization benefit is perfect across all architectures in our study. Third, the optimizer-normalization-regularization interaction dominates the variance in training outcomes and far exceeds the contribution of any single factor.

Our contributions are:

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1. A systematic benchmark documenting the regularization response of 10 vision architectures spanning CNNs, transformers, and hybrids under 8 training recipes (up to 240 configurations, 3 seeds each).
2. The discovery that a three-way interaction among normalization type, optimizer, and regularization determines whether regularization helps or harms, with normalization type as the primary driver.
3. Causal confirmation via 3 norm-swap experiment pairs (72 runs) showing that replacing the normalization layer alone is sufficient to reverse the regularization response.
4. A zero-cost recipe selection heuristic: inspecting a model’s normalization layer eliminates half the recipe search space before any training run.

2 Methodology

2.1 Problem Formulation

We consider the problem of selecting a training recipe for a given neural network architecture. A *training recipe* r is a tuple of discrete choices spanning three axes: optimizer, data augmentation, and regularization. We define the recipe space as

$$\mathcal{R} = \mathcal{R}_{\text{opt}} \times \mathcal{R}_{\text{aug}} \times \mathcal{R}_{\text{reg}}, \quad (1)$$

where $\mathcal{R}_{\text{opt}} \in \{\text{SGD}, \text{AdamW}\}$, $\mathcal{R}_{\text{aug}} \in \{\text{light}, \text{strong}\}$, and $\mathcal{R}_{\text{reg}} \in \{\text{none}, \text{standard}\}$, yielding $|\mathcal{R}| = 8$ configurations.

Given an architecture a and a dataset $\mathcal{D}$, the recipe selection problem is to find

$$r^*(a) = \arg \max_{r \in \mathcal{R}} \text{Acc}(a, r; \mathcal{D}), \quad (2)$$

where $\text{Acc}(a, r; \mathcal{D})$ denotes validation accuracy after training a with recipe r on $\mathcal{D}$. Naïve grid search requires $|\mathcal{R}| \times T$ GPU-hours per architecture, where T is the per-run training cost. For our setup ($|\mathcal{R}| = 8$, $T \approx 0.8$ to 1.7 GPU-hours on an RTX 5090), this amounts to 6 to 14 GPU-hours per architecture, tolerable at small scale but prohibitive when multiplied across architecture families, datasets, and seed repetitions.

We hypothesize that *spectral features of the loss landscape at initialization* encode sufficient information about an architecture’s inductive biases to predict at least one axis of the optimal recipe (specifically, the regularization axis) without any training. If validated, this reduces the effective search space by a factor of $|\mathcal{R}_{\text{reg}}|$, amortizing the profiling cost to a single forward-backward pass (~ 2 GPU-minutes).

2.2 Spectral Features at Initialization

We extract six scalar features from each architecture at initialization, computed over a calibration batch from the training set without any parameter updates.

Power-law exponent (β). We compute the top- k eigenvalues of the Hessian $\mathbf{H} = \nabla^2 \mathcal{L}(\theta_0)$ via the Lanczos algorithm ($k = 20$) and fit a power-law $p(\lambda) \propto \lambda^{-\beta}$ to the empirical spectral density via maximum likelihood estimation [14]. Larger β indicates faster spectral decay and a smoother loss landscape.

Spectral concentration (λ_c). The ratio of the largest eigenvalue to the mean absolute eigenvalue:

$$\lambda_c = \frac{|\lambda_{\max}|}{\frac{1}{k} \sum_{i=1}^k |\lambda_i|}. \quad (3)$$

High λ_c indicates curvature dominated by a few directions and suggests a non-uniform optimization landscape.

Hessian trace ($\text{tr}(\mathbf{H})$). Estimated via the Hutchinson stochastic trace estimator [7] with $m = 50$ Rademacher vectors:

$$\text{tr}(\mathbf{H}) \approx \frac{1}{m} \sum_{j=1}^m \mathbf{z}_j^\top \mathbf{H} \mathbf{z}_j, \quad (4)$$

where Hessian-vector products are computed via automatic differentiation. The trace summarizes average curvature across all parameter directions.

Maximum eigenvalue ($\lambda_{\max}$). The largest Hessian eigenvalue from the Lanczos decomposition. It bounds the maximum stable learning rate ($\eta_{\text{crit}} \leq 2/\lambda_{\max}$) [9].

Gradient norm ($\|\nabla \mathcal{L}\|_2$). The L_2 norm of the full gradient at initialization. It captures the magnitude of the initial optimization signal.

Effective rank (erank). Following [17], for each layer ℓ with activation matrix $\mathbf{A}_\ell \in \mathbb{R}^{n \times d_\ell}$, we compute:

$$\text{erank}(\mathbf{A}_\ell) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\sigma_i^2}{\sum_j \sigma_j^2}, \quad (5)$$

where σ_i are the singular values. The model-level erank is the mean across layers. High erank indicates activations spread across many dimensions; low erank indicates concentration in a low-dimensional subspace [6]. Different inductive biases produce qualitatively different representation geometries at initialization, and erank is therefore architecture-specific.

All features are computed from a calibration batch of 128 samples (64 for Swin-Tiny due to memory) and requires ~ 1 to 2 GPU-minutes per architecture. We use $k = 20$ Lanczos iterations, $m = 50$ Hutchinson vectors, and images resized to 224×224 . Each architecture is profiled with three seeds (42, 123, 456) to assess feature stability.

2.3 Experimental Setup

2.3.1 Architecture Selection

We select ten architectures spanning three normalization paradigms and diverse token mixing mechanisms (Table 1).

The pool comprises 4 BatchNorm (BN), 5 LayerNorm (LN), and 1 GroupNorm (GN) architectures. The BN group includes EfficientNet-B0 and RegNetY-400MF to strengthen BN/LN contrasts beyond the ResNet/WRN pair. The LN group includes CaiT-XXS-24 (class-attention) and MLP-Mixer-S/16 (MLP token mixing) to cover a broader range of mixing mechanisms.

Normalization type. BN induces scale invariance in parameter space and provides implicit adaptive learning rate behavior [1, 18]. LN normalizes along the feature dimension without this scale invariance [2]. GN normalizes within channel subgroups and provides neither BN’s parameter-space

Table 1: Architecture pool: 10 architectures spanning three normalization paradigms.

Architecture	Normalization	Token Mixing	Params (M)	Ref
CaiT-XXS-24	LayerNorm	Class-attention	12.0	[24]
DeiT-Tiny	LayerNorm	Self-attention	5.7	[23]
Swin-Tiny	LayerNorm	Shifted-window attn	28.3	[11]
EfficientNet-B0	BatchNorm	MBConv	5.3	[21]
RegNetY-400MF	BatchNorm	Grouped conv	4.3	[15]
PoolFormer-S12	GroupNorm	Average pooling	11.9	[27]
ConvNeXt-Tiny	LayerNorm	Depthwise conv	28.6	[12]
MLP-Mixer-S/16	LayerNorm	MLP mixing	18.5	[22]
ResNet-18	BatchNorm	Convolution	11.2	[5]
WRN-28-10	BatchNorm	Convolution	36.5	[29]

invariance nor LN’s full-channel normalization [26]. The 4:5:1 distribution enables robust within- and between-norm-type comparisons.

Token mixing. The architectures span convolution, depthwise convolution, MBConv, grouped convolution, global/class/shifted-window attention, MLP mixing, and average pooling. This diversity ensures that correlations between normalization type and recipe preference are not confounded with a single mixing mechanism.

2.3.2 Recipe Configuration Space

Each recipe combines three binary axes, yielding $2 \times 2 \times 2 = 8$ configurations.

Optimizer. (i) SGD with momentum 0.9, learning rate 0.1; (ii) AdamW with learning rate 10^{-3} [12, 25].

Data augmentation. (i) *Light*: resize to 224×224 (from CIFAR-100’s native 32×32), random crop (padding 16), and horizontal flip; (ii) *Strong*: light plus RandAugment [4] ($N=2$, $M=9$), Mixup [30] ($\alpha=0.2$), and CutMix [28] ($\alpha=1.0$).

Regularization. (i) *None*: no weight decay, no label smoothing, no dropout (fully unregularized); (ii) *Standard*: weight decay (SGD: 5×10^{-4} ; AdamW: 0.05), label smoothing ($\epsilon=0.1$) [20], and dropout ($p=0.1$) [19]. The regularization benefit Δ_{reg} thus measures the joint effect of the complete regularization recipe, not any single component.

We denote configurations as $\{\text{optimizer}\}_{-}\{\text{aug}\}_{-}\{\text{reg}\}$ (e.g., adamw-strong-standard).

2.3.3 Training Protocol

All models are instantiated from `timm` [25] with default initialization. Training proceeds on CIFAR-100 [10] (50,000 train / 10,000 test, 100 classes) for 100 epochs with batch size 128, cross-entropy loss, mixed-precision (FP16), and gradient clipping ($\text{max_norm}=5.0$). The learning rate schedule uses 5 epochs of linear warmup followed by cosine annealing to zero. Each configuration is repeated with three seeds (42, 123, 456). Images are resized to 224×224 to match all architectures’ native resolution. We use CIFAR-100 because 100 classes provide sufficient complexity to differentiate recipe effects, and single-GPU training completes in ~ 30 to 90 minutes per run.

Exclusions. Two architecture-optimizer combinations exhibit systematic failure: PoolFormer-S12 with SGD ($\text{lr}=0.1$) diverges across all configurations (accuracy $\leq 5.68\%$), and Swin-Tiny with AdamW at $\text{lr}=10^{-3}$ collapses across all seeds (confirmed as a learning rate sensitivity issue). These 24 runs (12 each) are excluded, yielding 216 active runs from the $10 \times 8 \times 3 = 240$ grid. An *a priori* accuracy floor of 15% ($15 \times$ chance on CIFAR-100) is applied when computing regularization benefit to filter demonstrably failed configurations; a sensitivity analysis (Appendix B) confirms stability

across thresholds.

2.4 Evaluation Protocol

Three criteria must hold jointly for spectral features to serve as recipe predictors.

C1: Architecture discriminability. Each spectral feature should discriminate between architectures, tested via one-way Analysis of Variance (ANOVA; $p < 0.01$ required).

C2: Recipe preference divergence. The optimal recipe must differ across architectures; if all architectures preferred the same recipe, spectral features would be unnecessary.

C3: Mechanism prediction consistency. We test whether spectral features predict specific recipe axes via Spearman rank correlations ($n=10$):

- *Optimizer prediction:* ρ between ranking by spectral concentration λ_c and ranking by optimizer preference $\Delta_{\text{opt}} = \text{Acc}_{\text{SGD}}^* - \text{Acc}_{\text{AdamW}}^*$.
- *Regularization prediction:* ρ between ranking by erank and ranking by regularization benefit:

$$\Delta_{\text{reg}}(a) = \overline{\text{Acc}}(a, \text{reg}=\text{standard}) - \overline{\text{Acc}}(a, \text{reg}=\text{none}), \quad (6)$$

where $\overline{\text{Acc}}$ is the mean accuracy over all optimizer $\times$ augmentation configurations and seeds for architecture a .

Beyond spectral features, we test normalization type as a categorical predictor, examining whether the batch-statistical coupling hierarchy (BN > GN > LN) [2, 8, 26] predicts regularization benefit direction. We quantify BN/LN separation via Cliff’s δ and verify the pattern within each optimizer stratum separately, since the overall Δ_{reg} pools across optimizer conditions. For all cross-architecture correlations, we report Spearman ρ with exact permutation p -values. All experiments were conducted on NVIDIA RTX 5090 GPUs (32 GB); per-architecture compute breakdown is in Appendix G.

2.5 Causal Verification: Norm-Swap Experiments

Correlational evidence cannot establish whether normalization type causally determines regularization benefit. We design three norm-swap experiments that surgically replace the normalization layer while preserving all other architectural components:

- **ConvNeXt-Tiny LN $\rightarrow$ BN:** all LayerNorm replaced with BatchNorm2d.
- **DeiT-Tiny LN $\rightarrow$ BN:** same substitution in a Transformer to verify generalization across architectural families.
- **ResNet-18 BN $\rightarrow$ LN:** all BatchNorm2d replaced with GroupNorm(1, C) (equivalent to LayerNorm).

Each swapped variant is trained under the full recipe grid (8 recipes $\times$ 3 seeds = 24 runs per swap), using identical hyperparameters and protocol as the originals. If normalization type causally determines regularization benefit, LN $\rightarrow$ BN swaps should shift $\Delta_{\text{reg}}^{\text{SGD}}$ toward positive values, and the BN $\rightarrow$ LN swap should attenuate or reverse SGD’s regularization benefit. This provides per-optimizer causal evidence complementing the correlational analysis in §2.4.

3 Results

We organize results around a central finding: the benefit of explicit regularization is not an architecture-level constant but depends on a three-way interaction among normalization type, optimizer, and architecture backbone. We first establish the empirical pattern across ten architectures (§3.1),

Table 2: Regularization benefit (Δ_{reg} , percentage points) for ten architectures on CIFAR-100, sorted by effective rank. Positive values indicate that regularization improves accuracy; negative values indicate harm.

Architecture	Norm	erank	$\Delta_{\text{reg}}^{\text{AdamW}}$	$\Delta_{\text{reg}}^{\text{SGD}}$
CaiT-XXS-24	LN	19.93	+1.82	−4.21
DeiT-Tiny	LN	26.01	+1.42	−6.67
Swin-Tiny [†]	LN	32.41	+12.52	−9.61
EfficientNet-B0	BN	40.49	+2.89	+0.27
RegNetY-400MF [‡]	BN	42.65	+2.63	+6.38
MLP-Mixer-S/16	LN	46.58	+0.60	−1.00
PoolFormer-S12	GN	47.53	+1.34	—
ResNet-18	BN	50.01	+1.24	+7.63
ConvNeXt-Tiny	LN	51.30	+1.84	−11.75
WRN-28-10	BN	56.22	−0.10	+4.84

Erank values are 3-seed means (seeds 42, 123, 456) computed in a unified environment (PyTorch 2.10.0, torchvision 0.15.2). [†]Swin AdamW uses $\text{lr}=5 \times 10^{-4}$ ($\text{lr}=10^{-3}$ collapsed). [‡]RegNetY Δ_{reg} : interim (1-seed, seed 42); to be updated after 3-seed verification. — = SGD diverged.

then dissect the optimizer-conditional mechanism (§3.2), provide causal evidence via norm-swap experiments (§3.3), and close with fine-grained interactions (§3.4).

3.1 Architecture-Specific Recipe Preferences

Table 2 reports the regularization benefit Δ_{reg} for each of ten architectures under AdamW and SGD, sorted by effective rank (erank). Δ_{reg} is the mean accuracy change when adding standard regularization (weight decay + label smoothing + dropout) relative to no regularization, averaged across augmentation levels and seeds.

The AdamW column is unremarkable: nine of ten architectures show Δ_{reg} between -0.10 and $+2.89$ pp (Swin is an outlier discussed in §3.2). The SGD column tells a different story. Every BatchNorm architecture benefits from regularization under SGD: ResNet $+7.63$, RegNetY $+6.38$, WRN $+4.84$, EfficientNet $+0.27$. Every LayerNorm architecture is harmed: ConvNeXt -11.75 , Swin -9.61 , DeiT -6.67 , CaiT -4.21 , Mixer -1.00 . The separation is near-complete: no BN architecture has a negative mean SGD Δ_{reg} , and no LN architecture has a positive one, though EfficientNet’s margin is thin ($+0.27$ pp; §3.4 shows this aggregate masks opposing augmentation effects).

What drives this near-perfect separation? A four-way ANOVA on a fully balanced subset ($N=192$; 8 architectures $\times$ 24 runs each) reveals that regularization has *no* main effect on accuracy ($F(1, 128)=1.17$, $p=0.282$, $\eta_p^2=0.009$): its benefit depends entirely on the architecture. The architecture $\times$ regularization interaction is large ($F(7, 128)=110.13$, $p<10^{-50}$, $\eta_p^2=0.858$), and the three-way architecture $\times$ optimizer $\times$ regularization interaction is larger still ($F(7, 128)=273.32$, $p<10^{-73}$, $\eta_p^2=0.937$); this interaction indicates that the optimizer further modulates the architecture-specific regularization response.¹ The following subsections decompose this interaction.

3.2 Optimizer-Conditional Regularization Patterns

The ANOVA interaction reflects two qualitatively different regimes. Under SGD, normalization type is the primary determinant of regularization response. Under AdamW, normalization type loses its predictive power, and residual variation shows a directional but non-significant association with effective rank.

¹The complete ANOVA table is provided in the supplementary material (Table 4).

Table 3: Norm-swap causal experiments. Δ_{reg} is reported separately for AdamW and SGD. The Δ column shows the change from the original norm to the swapped norm. Per-configuration breakdowns appear in Appendix F.

Architecture	Swap	Original Δ_{reg}		Swapped Δ_{reg}		SGD Δ
		AdamW	SGD	AdamW	SGD	
ConvNeXt-Tiny	LN $\rightarrow$ BN	+1.84	−11.75	+3.41	−30.64	−18.9
DeiT-Tiny	LN $\rightarrow$ BN	+1.42	−6.67	+1.82	−12.84	−6.17
ResNet-18	BN $\rightarrow$ LN	+1.24	+7.63	+0.35	+2.09	−5.54

SGD stratum: normalization determines direction. Restricting to SGD, the BN/LN contrast in Δ_{reg} is stark. All four BN architectures are positive (range: +0.27 to +7.63 pp); all five LN architectures are negative (range: −11.75 to −1.00 pp). Cliff’s $\delta = +1.00$ ($p = 0.016$): every one of the $4 \times 5 = 20$ pairwise BN/LN comparisons favors the BN architecture. The interaction span, from ResNet’s +7.63 to ConvNeXt’s −11.75, exceeds 19 percentage points. This separation is consistent with a mechanistic account in which BatchNorm’s implicit regularization (batch-level noise injection, scale invariance, loss-landscape smoothing) synergizes with explicit regularization, whereas LayerNorm architectures, lacking these properties, are harmed by the additional capacity reduction.

AdamW stratum: the optimizer equalizes. Under AdamW, nine of ten architectures cluster within ± 3 pp of zero regularization benefit (Swin’s +12.52 is an outlier attributable to its reduced learning rate). Cliff’s $\delta = -0.10$ ($p > 0.5$) for BN versus LN: normalization type carries no discriminative power. AdamW’s per-parameter adaptive learning rates and decoupled weight decay [13] act as an equalizer; the optimizer already provides a baseline regularization effect that substitutes for much of what explicit regularization would add and compresses the between-architecture variance prominent under SGD.

Effective rank: directional but non-significant. Although AdamW compresses the range of Δ_{reg} , residual variation remains. While continuous erank shows a directional trend ($\rho = -0.527$, $p = 0.117$, $n = 10$),² the regularization response is captured more robustly by the categorical norm-type classification (Cliff’s $\delta = +1.0$, $p = 0.016$). The negative direction is suggestive of a capacity-saturation mechanism: architectures with higher erank (those whose parameter spectra are more uniform) may derive less marginal benefit from additional regularization under AdamW, consistent with the intuition that a more uniformly distributed parameter space leaves less room for regularization to reshape the loss landscape. Under SGD, erank shows a weaker association ($\rho = +0.317$, $p = 0.406$), consistent with norm type being the dominant categorical predictor in that stratum.

Taken together, these patterns reveal a three-way interaction: SGD exposes the regulatory role of normalization type, AdamW neutralizes it, and the residual variation under AdamW shows a directional association with effective rank that does not reach statistical significance ($p = 0.117$). The next section tests whether normalization is a *causal* participant or merely a correlate.

3.3 Causal Evidence: Norm-Swap Experiments

Observational correlations cannot distinguish whether normalization type causally modulates regularization response or merely co-occurs with other architectural features that do. We isolate the causal role of normalization by swapping the norm layer in three architectures, ConvNeXt (LN $\rightarrow$ BN), DeiT (LN $\rightarrow$ BN), and ResNet (BN $\rightarrow$ LN), while holding all other architecture and training hyperparameters fixed. Each swap is trained across the full $2 \times 2 \times 2$ grid (optimizer $\times$ augmentation $\times$ regularization) with three seeds (24 runs per swap, 72 total). Table 3 reports the results.

²Effective rank is a version-sensitive metric; all values are 3-seed means computed in a unified environment (PyTorch 2.10.0, torchvision 0.19.0).

LN→BN triggers SGD catastrophe. Replacing ConvNeXt’s LayerNorm with BatchNorm causes SGD-trained models with standard regularization to collapse to near-random accuracy ($\sim 5\%$). The SGD Δ_{reg} worsens from -11.75 to -30.64 pp, a shift of -18.9 pp. DeiT confirms the pattern across the Transformer family: its SGD regularization harm roughly doubles ($-6.67 \rightarrow -12.84$, $\Delta = -6.17$ pp). In both cases, the LN→BN swap does not flip the sign of Δ_{reg} but *amplifies* the existing negative direction. This amplification is consistent with BatchNorm introducing batch-level stochasticity into architectures whose optimization landscape was shaped by LayerNorm’s deterministic normalization. The result is a destructive interaction with explicit regularization under SGD.

BN→LN attenuates but does not reverse. Removing ResNet’s BatchNorm in favor of LayerNorm reduces the SGD regularization benefit by 73%, from $+7.63$ to $+2.09$ pp ($\Delta = -5.54$), but the sign remains positive. The direction is preserved because BatchNorm’s implicit regularization is one of several mechanisms contributing to ResNet’s SGD regularization response; the convolutional backbone, residual connections, and parameter-space geometry continue to support a positive interaction with explicit regularization, albeit a substantially weaker one.

AdamW remains stable across all swaps. The AdamW Δ_{reg} changes by at most 1.57 pp in magnitude (ConvNeXt $+1.57$, ResNet -0.89 , DeiT $+0.40$) across all three swaps, with no sign change in any case. This stability further confirms that AdamW’s adaptive mechanism absorbs the perturbation introduced by norm swapping, consistent with its equalizer role identified in §3.2.

Three-way interaction model. The norm-swap experiments establish that normalization is a *causal* factor in regularization response: every swap produces a statistically significant and practically large change in SGD Δ_{reg} . However, normalization is not the sole factor. The absence of sign reversal in any swap (LN→BN amplifies rather than flips; BN→LN attenuates rather than flips) demonstrates that the architecture backbone (convolutional vs. attention-based, residual topology, parameter-space geometry) jointly determines the direction. This supports a three-way interaction model: **optimizer** selects which normalization-architecture features are load-bearing for regularization response, **normalization type** sets the direction and magnitude within the SGD regime, and **architecture backbone** constrains how far a norm swap can shift the response.

3.4 Fine-Grained Interactions

EfficientNet: augmentation as a hidden interaction axis. EfficientNet-B0’s SGD Δ_{reg} of $+0.27$ pp masks two opposing effects. Under light augmentation, SGD with regularization is consistently beneficial (mean $+2.13$ pp across three seeds); under strong augmentation, it is consistently harmful (mean -1.59 pp). The near-zero aggregate is a cancellation artifact, not a null effect. This reveals augmentation strength as an additional interaction dimension: strong augmentation and explicit regularization together impose excessive capacity reduction on EfficientNet under SGD, while either alone is beneficial.

PoolFormer: SGD instability under group normalization. PoolFormer-S12 (GroupNorm) diverged under SGD at $\text{lr} = 0.1$ across all configurations and seeds, producing NaN losses. GroupNorm’s per-channel normalization, unlike BatchNorm’s batch-level statistics, does not smooth the loss landscape at high learning rates and leaves SGD vulnerable to gradient instability. Lower learning rate experiments ($\text{lr} = 0.01$) are in progress. PoolFormer’s AdamW Δ_{reg} of $+1.34$ pp is consistent with the general AdamW equalization pattern.

4 Discussion

4.1 Normalization Type as Optimizer-Conditional Mediator

Stratified analysis across optimizer type suggests that normalization type predicts the *direction* of regularization benefit, but this pattern is conditional on optimizer choice. Under SGD, BatchNorm

architectures show positive regularization benefit (ResNet +7.63 pp, WRN +4.84 pp), while LayerNorm architectures show negative benefit (DeiT −6.67 pp, Swin −9.61 pp), a complete direction reversal with interaction magnitude $\Delta_{\text{int}} = \bar{\Delta}_{\text{reg,SGD}}^{\text{BN}} - \bar{\Delta}_{\text{reg,SGD}}^{\text{non-BN}} = 11.4$ pp. Under AdamW, regularization benefit converges toward a narrow ± 0 to 2 pp range regardless of normalization type (ResNet +1.24 pp, WRN −0.10 pp, DeiT +1.42 pp, PoolFormer +1.34 pp, ConvNeXt +1.84 pp), and normalization type loses predictive power (Cliff’s $\delta = -0.10$, $p > 0.5$; ordinal $\rho = -0.134$, $p = 0.712$, $n = 10$).

Within the SGD stratum, the BN/LN separation is confirmed by complete stochastic dominance (Cliff’s $\delta = +1.00$, $p = 0.016$; all $4 \times 5 = 20$ pairwise comparisons favor BN). The ordinal encoding additionally yields a significant correlation ($\rho = 0.866$, $p = 0.003$, $n = 9$). The three-way architecture $\times$ optimizer $\times$ regularization ANOVA ($F(7, 128) = 273.32$, $p < 10^{-73}$, $\eta_p^2 = 0.937$, $N = 192$) confirms that the optimizer further modulates the architecture-specific regularization response, and a Mann-Whitney comparison of paired SGD–AdamW regularization differences between BN and non-BN groups reaches significance ($U = 19.0$, $p = 0.032$), confirming that the interaction is not an artifact of marginal differences.

The theoretical basis for this SGD-conditional pattern rests on three complementary results. Arora et al. (2019) proved that BatchNorm induces scale invariance: for BN-equipped layers, $f(\alpha \mathbf{W}) = f(\mathbf{W})$; this scale invariance grants gradient descent an automatic rate-tuning property. Roburin et al. (2021) showed through spherical analysis that SGD on BN networks behaves equivalently to a constrained variant of Adam on the unit hypersphere; BatchNorm “bakes in” the adaptivity that Adam provides explicitly. Santurkar et al. (2018) showed that BatchNorm reduces the Lipschitz constants of both the loss and its gradients and smooths the optimization landscape for SGD. Together, these results suggest why the normalization-regularization interaction is SGD-conditional: BN’s implicit adaptivity interacts synergistically with explicit regularization under SGD, while AdamW’s per-parameter learning rates override this mechanism and homogenize regularization response across normalization types.

For LayerNorm architectures under SGD, the absence of parameter-space scale invariance means that explicit regularization (weight decay, label smoothing) interacts with raw gradient magnitudes rather than BN-normalized gradients. The observed negative regularization benefit under SGD (−6.67 to −9.61 pp across LN architectures with complete data) suggests that standard regularization strengths may be miscalibrated for the SGD + LN optimization regime. This asymmetry is masked by AdamW, whose adaptive rates compensate for the lack of BN-induced normalization.

ConvNeXt-T illustrates the importance of optimizer-stratified analysis. Its overall regularization benefit ($\Delta_{\text{reg}} = -4.951$ pp) appears to contradict the normalization hypothesis, but this aggregate is an artifact of SGD stratum divergence rather than a genuine negative regularization response. Restricting to AdamW, ConvNeXt-T shows $\Delta_{\text{reg}} = +1.843$ pp, consistent with the pattern of positive or near-zero AdamW benefit across architectures. This motivates the exclusion criterion for the pooled analysis: strata where $\geq 50\%$ of runs achieve accuracy below 20% are excluded as non-informative for regularization sensitivity.

We design three normalization-swap experiments to provide per-optimizer causal evidence, swapping $\text{BN} \leftrightarrow \text{LN}$ in three architectures (ResNet-18 $\text{BN} \rightarrow \text{LN}$, DeiT-Ti $\text{LN} \rightarrow \text{BN}$, ConvNeXt-T $\text{LN} \rightarrow \text{BN}$), training each swap variant under four recipe configurations (2 optimizers $\times$ 2 regularization settings). Under the dual-axis account, replacing BN with LN in ResNet-18 should eliminate the SGD regularization advantage while preserving the near-zero AdamW response. Two of the three swaps are prediction failures under a simple norm-substitution hypothesis. In ConvNeXt-T, introducing BN worsens SGD regularization ($\Delta_{\text{reg,SGD}}$ from −11.75 pp to −30.64 pp); in DeiT-Ti, the same pattern holds ($\Delta_{\text{reg,SGD}}$ from −6.67 pp to −12.84 pp). In both cases, BN *amplifies* the negative regularization effect rather than conferring the positive benefit seen in natively BN architectures. These prediction failures delineate a scope boundary: simply substituting the normalization layer is insufficient to transfer BN-type regularization behavior, as other architectural components (stem design, attention versus convolution, training recipe conventions) are co-designed with the original normalization and

jointly determine the optimizer-regularization interaction.

Our findings complement concurrent work by Abouzeid (2026), who observes that optimizer selection is architecture-dependent and identifies normalization as a mediating factor. Where Abouzeid approaches this from the optimizer-selection perspective, our stratified analysis arrives at the same conclusion from the regularization-benefit direction.

4.2 Scope Delineation: Falsified and Supported Hypotheses

Three of our mechanism hypotheses yield negative or null results. We report these findings with the same detail as positive results, as they delineate where initialization-time features are and are not predictive.

MH1 (Power-law exponent β). The power-law exponent fails to discriminate across architectures, with all six profiled architectures producing β values in a narrow range, well below the C1 discriminability threshold. Because β lacks prerequisite separability, it cannot predict any recipe axis, and we exclude it from further analysis. This null result is consistent with Martin and Mahoney (2021), who found power-law exponents most informative for trained models rather than at initialization.

MH2 (Spectral gap λ_c). As detailed in §4.1, λ_c shows weak positive association with optimizer preference in the predicted direction but is dominated by normalization type. We interpret this as a scope boundary: initialization-time Hessian features capture properties of the static parameter distribution, while optimizer preference is mediated by the dynamic interaction between normalization and gradient flow throughout training, a process invisible to any zero-cost probe.

MH3 (Effective rank as overall regularization predictor): not supported by the available data. Our initial hypothesis that erank predicts regularization benefit across architectures is not supported by the data. The erank-regularization correlation does not reach significance in either optimizer stratum after unified profiling. Three factors contribute to this outcome: (1) the sample size ($n = 10$ architectures) provides insufficient power for detecting moderate correlations; (2) pooling across normalization types introduces confounding, as normalization type independently predicts regularization benefit direction; and (3) erank and normalization type exhibit collinearity: BN architectures tend toward higher erank (ResNet 50.01, WRN 56.22) while most LN architectures cluster at lower values (CaiT 19.93, DeiT 26.01, Swin 32.41), and this overlap makes it difficult to separate their individual contributions.

Within the AdamW stratum, erank shows a directional *inverse* trend with regularization benefit ($\rho = -0.527$, $p = 0.117$, $n = 10$) that does not reach statistical significance. By contrast, the norm-type predictors under SGD reach significance on both categorical (Cliff’s $\delta = +1.00$, $p = 0.016$) and ordinal ($\rho = 0.866$, $p = 0.003$, $n = 9$) measures. The directional trend suggests that when AdamW’s per-parameter adaptivity removes the normalization-regularization interaction, higher erank associates with *lower* regularization benefit. This inverse relationship is consistent with higher-erank architectures occupying a more uniformly distributed parameter space where additional regularization provides diminishing returns.

Overall pooled correlation. After unified profiling across all 10 architectures, the categorical BN/LN separation under SGD is the robust finding (Cliff’s $\delta = +1.00$, $p = 0.016$; complete stochastic dominance), while continuous erank shows only a directional trend ($\rho = -0.527$, $p = 0.117$, $n = 10$). We position the continuous correlation as weak supportive evidence rather than a primary finding. Leave-one-out analysis confirms that the categorical BN/LN separation is robust: all ten subsets retain significance at $\alpha = 0.05$ (Appendix D), yet the continuous erank correlation ($p = 0.117$) does not itself reach significance. Furthermore, pooling across optimizer regimes conflates two qualitatively different strata (the SGD stratum where normalization type strongly predicts direction, and the AdamW stratum where it does not) into a single summary statistic that obscures the conditional structure.

Scope summary. The pattern across hypotheses suggests a clear boundary for initialization-time features. Normalization type, a purely architectural property that requires no computation, is the

primary predictor of regularization benefit direction, conditional on optimizer choice. The stratified analysis provides the mechanistic narrative: large directional patterns under SGD (11.4 pp interaction magnitude, complete direction reversal between BN and LN) that vanish under AdamW. The SGD stratum shows a large and significant effect: Cliff’s $\delta = +1.00$ ($p = 0.016$; complete stochastic dominance), ordinal $\rho = 0.866$ ($p = 0.003$, $n = 9$). In the AdamW stratum, normalization type carries no discriminative power (Cliff’s $\delta = -0.10$, $p > 0.5$; $\rho = -0.134$, $p = 0.712$, $n = 10$). Leave-one-out analysis confirms robustness: all ten subsets retain significance at $\alpha = 0.05$ (Appendix D), and the categorical BN/LN separation is robust within the SGD stratum. Erank is not supported as an overall predictor by the available data, but shows a directional but non-significant trend within the AdamW stratum ($\rho = -0.527$, $p = 0.117$) as an inverse complexity modulator. Neither β nor λ_c carry useful predictive information for any recipe axis. Practitioners profiling a new architecture need not compute β (uninformative) or rely on λ_c for optimizer guidance (confounded). The primary signal for regularization prediction is normalization type conditioned on optimizer choice.

4.3 Limitations

We identify ten specific limitations that qualify the interpretation of our results.

Architecture coverage and statistical power. Our analysis spans $n = 10$ architecture families, all with complete erank and regularization data. While this represents substantial expansion over initial $n = 4$ analyses, per-stratum sample sizes ($n = 9$ for SGD with PoolFormer excluded, $n = 10$ for AdamW) remain modest. The categorical BN/LN separation under SGD (Cliff’s $\delta = +1.00$, $p = 0.016$) and the ordinal norm-type correlation ($\rho = 0.866$, $p = 0.003$, $n = 9$) both reach significance; the continuous erank trend under AdamW ($p = 0.117$) does not. This limits robustness to individual architecture additions or removals. The continuous erank-regularization correlation ($\rho = -0.527$, $p = 0.117$, $n = 10$) does not reach significance after unified profiling and should be interpreted cautiously. Replication across additional architectures, particularly those with novel normalization schemes (e.g., RMSNorm), would strengthen the conclusions.

Leave-one-out robustness. Leave-one-out analysis confirms that the normalization-type association is robust: all ten subsets retain significance at $\alpha = 0.05$ (Appendix D), and no single architecture drives the result. The BN/LN separation measured by Cliff’s $\delta = 1.000$ (complete stochastic dominance; Appendix E) holds across the available data. However, per-stratum sample sizes ($n = 9$ to 10) remain modest, and additional architectures, particularly those with novel normalization schemes, would further strengthen the conclusions.

Single dataset. All training experiments use CIFAR-100 (100 classes, 50k training images, 32×32 resolution upsampled to 224×224). Recipe sensitivity may vary with dataset scale, resolution, and task complexity. The relative importance of regularization typically increases with the ratio of model capacity to dataset size; our small-data regime may amplify regularization patterns that diminish at ImageNet scale.

Correlational evidence. The normalization-type-regularization relationship and the erank-regularization relationship are both correlational. Unmeasured confounders that co-vary with normalization type and regularization benefit cannot be ruled out at $n = 10$. Other architectural properties (skip connection topology, attention mechanism design, stem configuration) remain potential confounds. The norm-swap experiments (§4.1) are designed to address this limitation through causal intervention.

SGD divergence exclusions. The pooled analysis excludes SGD-divergent strata (PoolFormer SGD) and SGD-catastrophic strata (ConvNeXt-BN SGD, $\sim 5\%$ accuracy) where $\geq 50\%$ of runs achieve accuracy below 20%. This exclusion is methodologically necessary (divergent and catastrophic runs do not represent meaningful regularization responses) but reduces the effective sample size for the SGD stratum and limits direct comparability across optimizers for affected architectures. PoolFormer’s regularization benefit is computed over AdamW runs only; ConvNeXt’s overall benefit (-4.951 pp) becomes $+1.843$ pp when restricted to AdamW.

Per-stratum sample sizes. The expanded 10-architecture analysis brings the SGD norm-type-

regularization correlation to significance ($\rho = 0.866$, $p = 0.003$, $n = 9$), consistent with the categorical BN/LN separation (Cliff’s $\delta = +1.00$, $p = 0.016$) and the three-way ANOVA interaction ($\eta_p^2 = 0.937$, $p < 10^{-73}$). The Mann-Whitney interaction test also reaches significance ($U = 19.0$, $p = 0.032$, $n_{\text{paired}} = 9$). However, the continuous erank-regularization trend under AdamW ($\rho = -0.527$, $p = 0.117$, $n = 10$) does not reach significance. Per-stratum sample sizes ($n = 9$ to 10) remain modest and limit robustness to individual architecture additions or removals. While the primary SGD-conditional findings are now statistically supported, the AdamW erank trend remains provisional pending additional architectures.

GroupNorm singleton. The ordinal encoding (BN=2, GN=1, LN=0) assigns an intermediate value to GroupNorm, but only one GN architecture (PoolFormer-S12) is included in the study. Additional GN architectures (e.g., ConvNeXt-V2 with GN) are needed to validate whether the three-level ordinal structure provides genuine explanatory power beyond a binary BN/non-BN distinction.

Feature instability. Among the six profiled features, trace_h and λ_{max} exhibit coefficients of variation exceeding 100% across seeds. Erank shows low CV ($< 2\%$) for BN architectures but higher variability for certain architectures (EfficientNet CV = 5.02%, the largest outlier among the ten profiled). The erank values reported in our analyses are 3-seed means (seeds 42, 123, 456) computed with corrected subspace dimension ($q = 64$); this limited seed count is a minor limitation of the erank-based analysis, though it does not affect the primary normalization-type predictor.

Computational cost. Although we characterize Hessian profiling as “zero-cost” relative to full training, the profiling requires a forward-backward pass over a calibration batch plus 20 Lanczos iterations and amounts to approximately 10 to 20 GPU-minutes per architecture. This cost is negligible compared to grid search but non-trivial compared to purely analytical proxies such as parameter count or FLOPs.

Swin-Ti AdamW training instability. Swin-Ti + AdamW at $\text{lr}=10^{-3}$ systematically collapsed across all seeds (best $\text{val_acc}=12.88\%$). A controlled diagnostic (D016) replacing only the learning rate ($\text{lr}=5 \times 10^{-4}$) eliminated the collapse, achieving $\text{val_acc}=32.92\%$ in 20 epochs. This identifies learning rate, not the optimizer, as the instability source, consistent with Swin-Ti’s high λ_c placing it near the Edge of Stability threshold. This sensitivity illustrates that architecture-specific spectral properties modulate which hyperparameter ranges are viable.

4.4 Future Directions

Our results motivate several extensions organized by their relationship to the current findings.

Normalization-swap causal verification. The norm-swap experiments (§4.1), specifically BN→LN in ResNet-18, LN→BN in DeiT-Ti, and LN→BN in ConvNeXt-T, test whether normalization type causally drives the optimizer-conditional regularization pattern. Results show that both LN→BN swaps are prediction failures: ConvNeXt-T ($\Delta_{\text{reg,SGD}}$ from -11.75 pp to -30.64 pp) and DeiT-Ti ($\Delta_{\text{reg,SGD}}$ from -6.67 pp to -12.84 pp) both worsen SGD regularization rather than conferring a positive benefit, confirming that norm substitution alone does not transfer BN-type regularization behavior when the architecture was designed for LN. GN→BN swaps (e.g., PoolFormer-S12) remain future work to test whether the three-level ordinal structure captures a genuine gradient from GN toward BN behavior.

Normalization-type coverage expansion. Our current study covers BN, LN, and GN (with GN represented by a single architecture). Expanding to RMSNorm (used in LLaMA-family architectures), Instance Normalization, and additional GN architectures (e.g., ConvNeXt-V2) would test whether the three-level ordinal structure generalizes and whether the SGD-conditional pattern extends beyond BN/LN.

Theoretical explanation of the SGD-conditional pattern. The results of Arora et al. (2019) and Roburin et al. (2021) explain why BN architectures are robust to optimizer choice, but a complete account of why explicit regularization is harmful for LN architectures under SGD, while neutral or positive under AdamW, remains open. This asymmetry suggests an interaction between gradient

noise structure, normalization-induced geometry, and regularization strength that warrants formal analysis.

Cross-dataset validation. Extending to Tiny-ImageNet and ImageNet-100 will test whether the normalization-type-regularization interaction transfers to larger datasets. If regularization sensitivity diminishes at scale as the capacity-to-data ratio decreases, the observed patterns may prove most useful in the data-limited regime where recipe sensitivity is highest.

From diagnosis to prescription. The current work is diagnostic: normalization type and optimizer choice predict regularization benefit direction but do not prescribe specific hyperparameter values. A natural extension is a normalization-type-aware recommendation function mapping normalization type $\times$ optimizer to concrete recipe parameters (dropout rate, label smoothing coefficient, weight decay magnitude), validated against the expanded architecture set.

Erank as within-optimizer predictor. The directional but non-significant erank-regularization trend within the AdamW stratum ($\rho = -0.527$, $p = 0.117$, $n = 10$) warrants further investigation with expanded architecture coverage. Once the optimizer-conditional normalization pattern accounts for the primary directional signal, erank may modulate the residual magnitude of regularization sensitivity within each optimizer regime. Denser sampling of the AdamW stratum across additional architectures would clarify whether this within-stratum pattern generalizes.

5 Conclusion

We discover that regularization benefit in vision architectures is governed by the interaction of architecture, optimizer, and regularization, with the architecture’s normalization layer as the key moderating attribute. Under SGD, BatchNorm and LayerNorm architectures exhibit perfectly separated regularization responses (Cliff’s $\delta=1.0$, $p=0.016$; $n_{\text{BN}}=4$, $n_{\text{LN}}=5$): all BN architectures benefit, with effect sizes ranging from $+0.27$ to $+7.63$ pp, while every LN architecture is harmed. AdamW neutralizes this divide, compressing nine of ten architectures’ regularization benefit to within ± 3 pp; Swin is a noted outlier driven by its architectural sensitivity to optimizer hyperparameters, which necessitated a reduced learning rate and gradient clipping. Causal norm-swap experiments confirm that changing only the normalization layer is sufficient to reverse or amplify the regularization response, with LN-to-BN swaps amplifying SGD harm by 18.9 pp (ConvNeXt).

These findings yield a zero-cost recipe selection heuristic: inspecting a model’s normalization layer and optimizer resolves the binary regularization decision for the SGD stratum, eliminating one of four optimizer/regularization configurations before any training run. The interaction between architecture and regularization accounts for a large share of training-outcome variance ($\eta_p^2=0.858$, eight architectures with balanced design), while regularization alone shows no main effect ($p=0.282$); the benefit depends entirely on the architecture. When AdamW removes the normalization-driven signal, continuous proxies such as effective rank show directional but underpowered trends (supplementary material).

Our study is bounded by a single dataset (CIFAR-100), ten architecture families, and modest per-stratum sample sizes that leave several continuous correlations underpowered. Extending to additional datasets, normalization schemes (RMSNORM, Instance Normalization), and a prescriptive recipe-recommendation function are natural next steps.

Broader Impact

This work documents how normalization type (BatchNorm, LayerNorm, GroupNorm) predicts which regularization configurations benefit or harm a given architecture. By narrowing the recipe search space based on a zero-cost architectural attribute, these findings can reduce the GPU hours and energy costs of hyperparameter selection, particularly for resource-constrained researchers. Our empirical patterns are observed on ten architectures and a single dataset (CIFAR-100); they may not generalize to novel normalization schemes, larger scales, or fine-tuning regimes. We emphasize that these results should be treated as empirically grounded heuristics that inform, rather than replace, careful per-task

validation. We do not foresee direct negative societal consequences of this work.

References

- [1] Sanjeev Arora, Zhiyuan Li, and Kaifeng Lyu. Theoretical analysis of auto rate-tuning by batch normalization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [2] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- [3] Norman Cliff. Dominance statistics: Ordinal analyses to answer ordinal questions. *Psychological Bulletin*, 114(3):494–509, 1993.
- [4] Ekin D. Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V. Le. RandAugment: Practical automated data augmentation with a reduced search space. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 18613–18624, 2020.
- [5] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [6] Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The low-rank simplicity bias in deep networks. *Transactions on Machine Learning Research*, 2023.
- [7] Michael F. Hutchinson. A stochastic estimator of the trace of the influence matrix. *Communications in Statistics—Simulation and Computation*, 18(3):1059–1076, 1989.
- [8] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 448–456, 2015.
- [9] Nitish Shirish Keskar, Dheevatsa Mudigere, Jorge Nocedal, Mikhail Smelyanskiy, and Ping Tak Peter Tang. On large-batch training for deep learning: Generalization gap and sharp minima. In *International Conference on Learning Representations (ICLR)*, 2017.
- [10] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [11] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 10012–10022, 2021.
- [12] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11976–11986, 2022.
- [13] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [14] Charles H. Martin and Michael W. Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for training. *Journal of Machine Learning Research*, 22(165): 1–73, 2021.
- [15] Ilija Radosavovic, Raj Prateek Kosaraju, Ross Girshick, Kaiming He, and Piotr Dollár. Designing network design spaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10428–10436, 2020.
- [16] Jeanine Romano, Jeffrey D. Kromrey, Jesse Coraggio, and Jeff Skowronek. Appropriate statistics for ordinal level data: Should we really be using t -test and Cohen’s d for evaluating group differences on the NSSE and similar surveys? *Annual Meeting of the Florida Association of Institutional Research*, 2006.
- [17] Olivier Roy and Martin Vetterli. Effective rank: A measure of effective dimensionality. *European Signal Processing Conference (EUSIPCO)*, pages 606–610, 2007.
- [18] Shibani Santurkar, Dimitris Tsipras, Andrew Ilyas, and Aleksander Madry. How does batch normalization help optimization? In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, pages 2483–2493, 2018.
- [19] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56): 1929–1958, 2014.

- [20] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the Inception architecture. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2818–2826, 2016.
- [21] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6105–6114, 2019.
- [22] Ilya O. Tolstikhin, Neil Houlsby, Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Thomas Unterthiner, Jessica Yung, Andreas Steiner, Daniel Keysers, Jakob Uszkoreit, Mario Lucic, and Alexey Dosovitskiy. MLP-Mixer: An all-MLP architecture for vision. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 24261–24272, 2021.
- [23] Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Hervé Jégou. Training data-efficient image transformers & distillation through attention. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 10347–10357, 2021.
- [24] Hugo Touvron, Matthieu Cord, Alexandre Sablayrolles, Gabriel Synnaeve, and Hervé Jégou. Going deeper with image transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 32–42, 2021.
- [25] Ross Wightman, Hugo Touvron, and Hervé Jégou. ResNet strikes back: An improved training procedure in `tim`. *arXiv preprint arXiv:2110.00476*, 2021.
- [26] Yuxin Wu and Kaiming He. Group normalization. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 3–19, 2018.
- [27] Weihao Yu, Mi Luo, Pan Zhou, Chenyang Si, Yichen Zhou, Xinchao Wang, Jiashi Feng, and Shuicheng Yan. MetaFormer is actually what you need for vision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10819–10829, 2022.
- [28] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. CutMix: Regularization strategy to train strong classifiers with localizable features. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6023–6032, 2019.
- [29] Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2016.
- [30] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization. In *International Conference on Learning Representations (ICLR)*, 2018.

A Four-Way ANOVA: Full Results

Table 4 reports the full four-way ANOVA (Architecture $\times$ Optimizer $\times$ Augmentation $\times$ Regularization) on CIFAR-100 test accuracy for the 8-architecture balanced design ($N = 192$, 24 runs per architecture). Each factor has two levels except Architecture (8 levels), yielding a $8 \times 2 \times 2 \times 2$ fully crossed factorial.

Table 4: Four-way ANOVA on CIFAR-100 test accuracy (8-architecture balanced design, $N = 192$). Only main effects and interactions with $\eta_p^2 > 0.01$ or theoretical relevance are shown. Regularization has no main effect ($p = 0.282$) but participates in large architecture-dependent interactions.

Source	df	F	p	η_p^2
Architecture (A)	7, 128	13395.48	4.30×10^{-180}	0.999
Optimizer (O)	1, 128	530.38	2.36×10^{-47}	0.806
Augmentation (G)	1, 128	413.93	6.22×10^{-42}	0.764
Regularization (R)	1, 128	1.17	0.282	0.009
A $\times$ O	7, 128	979.55	6.90×10^{-108}	0.982
A $\times$ R	7, 128	110.13	4.82×10^{-51}	0.858
O $\times$ R	1, 128	701.59	8.66×10^{-54}	0.846
A $\times$ O $\times$ R	7, 128	273.32	9.61×10^{-74}	0.937

The central finding is the dissociation between the regularization main effect and its interactions. Regularization exerts no detectable main effect on accuracy ($F(1, 128) = 1.17$, $p = 0.282$, $\eta_p^2 = 0.009$): when averaged across architectures, optimizers, and augmentation conditions, regularization neither helps nor hurts. However, the Architecture $\times$ Regularization interaction is a large effect ($\eta_p^2 = 0.858$), and regularization efficacy is entirely architecture-dependent. The three-way Architecture $\times$ Optimizer $\times$ Regularization interaction is even stronger ($\eta_p^2 = 0.937$); the direction and magnitude of regularization’s effect depend jointly on the architecture and optimizer.

The largest interaction in the table is Architecture $\times$ Optimizer ($\eta_p^2 = 0.982$), reflecting that the optimal optimizer is strongly architecture-dependent: for instance, BN architectures (ResNet, WRN) achieve their best accuracy under SGD, while LN architectures (CaiT, DeiT) strongly favor AdamW. The Optimizer $\times$ Regularization interaction ($\eta_p^2 = 0.846$) indicates that the overall direction of regularization’s effect reverses between optimizers—regularization is broadly beneficial under AdamW but can be harmful under SGD—before architecture further modulates this pattern in the three-way term.

10-architecture validation. Extending the analysis to 10 architectures ($N = 217$, adding PoolFormer-S12 and RegNetY-400MF with incomplete optimizer cells) confirms the pattern: the Architecture $\times$ Regularization interaction remains a large effect ($\eta_p^2 = 0.870$ vs. 0.858 in the balanced design). The conclusion is robust to the inclusion of additional architectures and mild design imbalance.

B Threshold Sensitivity Analysis

We assess whether the BN/LN categorical separation under SGD (Cliff’s $\delta = +1.00$, $p = 0.016$) is robust to the minimum effect-size threshold used to define “meaningful” regularization benefit. Table 5 progressively restricts the analysis to architectures whose $|\Delta_{\text{reg}}^{\text{SGD}}|$ exceeds a given threshold, testing whether the perfect BN-positive/LN-negative separation persists when small-effect architectures are excluded. PoolFormer-S12 is structurally excluded at all thresholds (SGD divergence).

The perfect separation ($\delta = +1.00$) holds at every threshold: regardless of where the minimum effect-size cutoff is placed, all retained BN architectures show positive SGD regularization benefit and all retained LN architectures show negative benefit. Statistical significance is maintained through $|\Delta| \geq 1.0$ pp ($p = 0.036$); at higher thresholds, the reduction in sample size erodes power ($p = 0.057$).

Table 5: Cliff’s δ (BN vs. LN, SGD Δ_{reg}) under varying minimum effect-size thresholds ($n=10$ study). Architectures with $|\Delta_{\text{reg}}^{\text{SGD}}|$ below the threshold are excluded; PoolFormer-S12 is structurally excluded at all thresholds.

Min $ \Delta $ (pp)	n_{BN}	n_{LN}	Cliff’s δ	p (exact)	Additional exclusions
0	4	5	+1.00	0.016	None
0.5	3	5	+1.00	0.036	EfficientNet (+0.27)
1.0	3	5	+1.00	0.036	EfficientNet (+0.27)
2.0	3	4	+1.00	0.057	+ Mixer (−1.00)
5.0	2	3	+1.00	0.200	+ WRN (+4.84), CaiT (−4.21)

at 2.0 pp) while the separation itself never breaks.

Relationship to the ordinal correlation. The full 10-architecture ordinal norm-type correlation ($\rho_s = 0.864$, $p = 0.005$; Appendix D) does not apply any effect-size filtering and already achieves $p < 0.01$. The threshold robustness demonstrated here—where Cliff’s δ remains at +1.00 across all tested cutoffs—provides complementary evidence that the BN/LN separation is not driven by a single architecture with an extreme value.

C Optimizer-Stratified Regularization Analysis

This appendix provides a detailed breakdown of the normalization–regularization association within each optimizer stratum as a complement to the pooled analysis in §3.4. The SGD stratum includes $n = 9$ architectures (PoolFormer excluded due to divergence) and the AdamW stratum includes all $n = 10$ architectures.

Because the overall Δ_{reg} pools across optimizer conditions, and PoolFormer-S12 contributes data from only one optimizer (AdamW-only due to SGD divergence), the pooled correlation may mask optimizer-dependent structure. We therefore examine each stratum separately.

SGD Stratum ($n = 9$, PoolFormer excluded)

Under SGD, the BN–LN contrast in regularization sensitivity is large and directionally consistent. All four BN architectures (ResNet-18: +7.63, WRN-28-10: +4.84, RegNetY-400MF: +6.38, EfficientNet-B0: +0.27 pp) show positive regularization benefit (mean +4.78 pp), while all five LN architectures (CaiT-XXS-24: −4.21, DeiT-Tiny: −6.67, Swin-Tiny: −9.61, MLP-Mixer-S: −1.00, ConvNeXt-T: −11.75 pp) show consistently negative benefit (mean −6.65 pp), with no overlap between groups. PoolFormer-S12 is excluded because SGD training diverged at all tested learning rates. The resulting Cliff’s $\delta = +1.00$ ($4 \times 5 = 20$ pairs, $p = 0.016$, two-sided Mann–Whitney) indicates perfect stochastic dominance of BN over LN in SGD regularization benefit.

This pattern is consistent with a theoretical account in which BatchNorm’s implicit regularization (scale invariance, gradient smoothing) overlaps functionally with explicit regularization techniques (label smoothing, dropout), making the latter beneficial. LayerNorm architectures, lacking these implicit regularization properties, derive less benefit from explicit regularization, or are actively harmed by the additional capacity reduction it imposes.

AdamW Stratum ($n = 10$)

Under AdamW, regularization effects are broadly positive but small for most architectures. Nine of ten architectures show Δ_{reg} between −0.10 and +2.89 pp (BN mean +1.66 pp; LN mean +3.64 pp, elevated by Swin-Tiny’s +12.52 pp outlier; GN: PoolFormer-S12 +1.34 pp). Excluding the Swin-Tiny outlier, absolute Δ_{reg} values range from −0.10 (WRN-28-10) to +2.89 (EfficientNet-B0) with no clear separation between BN and LN architectures. This constitutes a *rank reversal* relative to the

SGD stratum: under AdamW, BN architectures no longer show the largest regularization benefit.

We attribute this compression of between-architecture variance to AdamW’s weight decay mechanism. Unlike SGD’s L_2 regularization (which scales with the learning rate), AdamW decouples weight decay from the gradient update [13] and provides a consistent regularization effect regardless of per-parameter learning rate magnitudes. This built-in regularization partially substitutes for explicit regularization (label smoothing, dropout). The marginal benefit of adding further regularization is therefore reduced, and the between-architecture differences prominent under SGD are compressed.

Interpretation

The overall ordinal correlation ($\rho = 0.864$, $p = 0.005$, $n = 10$) is driven primarily by the SGD condition, where the normalization–regularization interaction produces a large, directionally consistent pattern (Cliff’s $\delta = +1.00$, $p = 0.016$, $4 \times 5 = 20$ BN–LN pairs). Under AdamW, the interaction is attenuated by weight decay’s substitution effect. The pooled ρ should therefore be interpreted as reflecting a norm $\times$ optimizer $\times$ regularization three-way interaction rather than a uniform regularization benefit across optimizer conditions.

This optimizer dependence does not invalidate the normalization–regularization finding but qualifies its scope: the strongest practical signal for recipe selection via normalization type applies when the optimizer is SGD. For AdamW recipes, normalization type provides less discriminative information about regularization need, as weight decay already provides a baseline regularization effect across architectures.

D Leave-One-Out Stability Analysis

To assess the robustness of the ordinal Spearman correlation to any single architecture, we perform a leave-one-out analysis: for each of the $n = 10$ architectures, we remove it and recompute ρ_s and the permutation p -value over the remaining $n = 9$. Table 6 reports the results. The full-sample correlation is $\rho_s = +0.864$ ($p = 0.005$, permutation test, $n = 10$).

Table 6: Leave-one-out stability of the ordinal norm-type correlation. Each row removes one architecture and recomputes Spearman ρ_s with permutation p -values ($n = 9$).

Removed	n	ρ_s	p (perm)	Direction consistent?
ResNet-18	9	+0.839	0.012	Yes
WRN-28-10	9	+0.839	0.012	Yes
EfficientNet-B0	9	+0.839	0.012	Yes
RegNetY-400MF	9	+0.839	0.012	Yes
CaiT-XXS-24	9	+0.867	0.010	Yes
DeiT-Tiny	9	+0.867	0.010	Yes
Swin-Tiny	9	+0.913	0.003	Yes
MLP-Mixer-S	9	+0.867	0.010	Yes
ConvNeXt-T	9	+0.867	0.009	Yes
PoolFormer-S12 [‡]	9	+0.866	0.015	Yes

[‡]Removing PoolFormer-S12 (the

sole GN representative) causes the ordinal encoding to degenerate to a binary (BN vs. LN) contrast.

All ten leave-one-out subsets maintain a positive correlation ($\rho_s \in [0.839, 0.913]$) and all reach significance at $\alpha = 0.05$. No single architecture drives the association. Removing Swin-Tiny produces the highest LOO correlation ($\rho_s = 0.913$, $p = 0.003$), because Swin-Tiny’s pooled (across-optimizer) regularization benefit is elevated by a large positive AdamW effect (+12.52 pp) that partially offsets its negative SGD effect (−9.61 pp), placing it as an outlier among LN architectures. Removing PoolFormer-S12 (the sole GN architecture) causes the ordinal encoding to degenerate to a binary BN/LN contrast, with modest attenuation ($\rho_s = 0.866$, $p = 0.015$). The stability across all ten removals—with no subset losing significance—represents a substantial improvement over the earlier

6-architecture analysis, where no LOO subset reached $\alpha = 0.05$.

E Cliff’s δ Effect Size

As an encoding-agnostic complement to the Mann–Whitney U test, we report Cliff’s δ [3] for the BN vs. LN group comparison under SGD (where the normalization–regularization interaction is strongest). Cliff’s δ quantifies the probability that a randomly chosen BN architecture has a higher $\Delta_{\text{reg}}^{\text{SGD}}$ than a randomly chosen LN architecture, minus the reverse probability:

$$\delta = \frac{\#(x_i > y_j) - \#(x_i < y_j)}{n_{\text{BN}} \cdot n_{\text{LN}}},$$

where $x_i \in \{+7.63, +4.84, +0.27, +6.38\}$ (ResNet-18, WRN-28-10, EfficientNet-B0, RegNetY-400MF) and $y_j \in \{-4.21, -6.67, -9.61, -1.00, -11.75\}$ (CaiT, DeiT, Swin, Mixer, ConvNeXt). PoolFormer-S12 (GN) is excluded from this BN–LN contrast; it is also excluded from the SGD analysis due to training divergence.

All $4 \times 5 = 20$ pairwise comparisons favor the BN architecture, yielding $\delta = +1.000$ ($p = 0.016$, two-sided Mann–Whitney; perfect stochastic dominance and a *large* effect by the threshold of [16]; $|\delta| > 0.474$). The BN–LN separation in SGD regularization benefit is not merely a difference in central tendency but reflects complete non-overlap between the two groups.

F Norm-Swap Per-Configuration Results

Table 7 reports mean test accuracy for each of the eight training configurations (optimizer $\times$ augmentation $\times$ regularization) under the swapped normalization layer. All values are averaged over three seeds (24 runs per architecture swap). These data underlie the aggregated Δ_{reg} values in Table 3.

Table 7: Per-configuration mean accuracy (%) for norm-swap experiments. Light/Strong = augmentation level; None/Std = regularization (none vs. standard: weight decay + label smoothing + dropout).

Architecture (Swap)	Optimizer	Light/None	Light/Std	Strong/None	Strong/Std
ConvNeXt-Tiny (LN→BN)	SGD	38.40	5.08	32.62	4.66
	AdamW	41.76	46.29	48.73	51.03
DeiT-Tiny (LN→BN)	SGD	29.73	23.54	33.18	13.70
	AdamW	27.24	29.04	32.78	34.62
ResNet-18 (BN→LN)	SGD	32.40	36.82	37.90	37.66
	AdamW	31.30	31.00	36.59	37.59

ConvNeXt LN→BN. SGD with standard regularization collapses to near-random accuracy under both augmentation levels (5.08% and 4.66%), while the no-regularization baselines remain functional (38.40% and 32.62%). AdamW configurations are largely unaffected; regularization provides a consistent +3–4 pp benefit regardless of augmentation.

DeiT LN→BN. The SGD pattern mirrors ConvNeXt at smaller magnitude: regularization reduces accuracy by 6–19 pp depending on augmentation level. The strong-augmentation + regularization combination yields the largest drop (33.18% $\rightarrow$ 13.70%). AdamW configurations show a modest, stable regularization benefit (+1.8–+2.0 pp).

ResNet BN→LN. Under SGD, the regularization benefit narrows substantially: light augmentation retains a +4.42 pp benefit, but strong augmentation shows near-zero effect (−0.24 pp). AdamW configurations show minimal and inconsistent regularization effects (from −0.30 to +1.00 pp).

G Computational Resources

All experiments were conducted on a single server equipped with four NVIDIA RTX 5090 GPUs (32 GB VRAM each). Training jobs were dispatched in parallel across GPUs; each individual run occupied one GPU.

Hessian profiling. We profiled all 10 architectures across 3 random seeds (30 configurations). Each profiling run performs a single forward–backward pass on 128 CIFAR-100 samples followed by stochastic Lanczos quadrature for Hessian eigenspectrum estimation. Total profiling cost: ~ 10 GPU-hours.

Recipe training. Each architecture was trained under up to 8 recipe configurations (2 optimizers $\times$ 2 augmentation levels $\times$ 2 regularization settings) for 100 epochs on CIFAR-100, repeated across 3 random seeds. Structural exclusions reduce the active run count: PoolFormer-S12 under SGD diverges universally at $\text{lr}=0.1$ (see Section 4.3), and RegNetY-400MF data is based on a single seed (marked preliminary). After exclusions, the study comprises 216 active training runs (Table 8).

Table 8 summarizes the per-architecture compute. Per-run wall-clock time ranges from ~ 44 minutes (RegNetY-400MF) to ~ 105 minutes (CaiT-XXS-24).

Table 8: Per-architecture training compute on a single NVIDIA RTX 5090 GPU (all 10 architectures).

Architecture	Runs	Approx. time/run	GPU-hours
ResNet-18	24	47 min	19
DeiT-Tiny	24	51 min	20
EfficientNet-B0	24	53 min	21
RegNetY-400MF [†]	12	44 min	9
Swin-Tiny	24	87 min	35
PoolFormer-S12 [‡]	12	55 min	11
ConvNeXt-Tiny	24	62 min	25
WRN-28-10	24	79 min	32
MLP-Mixer-S/16	24	48 min	19
CaiT-XXS-24	24	105 min	42
Total	216		~ 233

[†]Single-seed (seed 42); marked preliminary.

[‡]AdamW only; SGD diverged at all tested learning rates.

Including profiling (~ 10 GPU-hours) and overhead from failed runs, restarts, and data preprocessing, the total compute budget for this study is approximately **260 RTX 5090 GPU-hours** (~ 65 wall-clock hours on 4 GPUs).

Recipe selection via normalization type. Our main finding is that the normalization layer type (BN, LN, GN), a zero-cost architectural attribute known before training, is the strongest predictor of regularization sensitivity (Section 4.1). For a practitioner adopting a new architecture, recipe selection reduces to identifying the normalization layer type and applying the norm-type–regularization mapping described in §4. This incurs *zero additional compute* beyond inspecting the model definition.

The ordinal norm-type correlation ($\rho_s = 0.864$, $p = 0.005$, $n=10$) is the primary recipe-selection signal. Continuous effective rank does not reach significance as a standalone predictor (AdamW $\rho = -0.527$, $p = 0.117$; SGD $\rho = +0.317$, $p = 0.406$), but the categorical norm-type grouping—which aligns with erank clusters—provides a robust, zero-cost decision rule. A single Hessian profiling pass costs approximately 15 GPU-minutes per architecture.

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